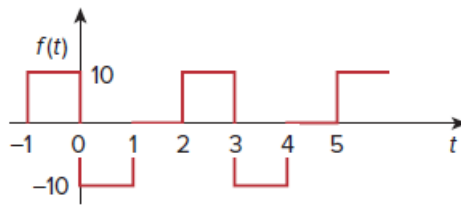


Problem

Find the rms value of the signal shown in Fig. 11.56.

Figure 11.56:



Step-by-step solution

Step 1 of 2

Refer to Figure 11.56 in the text book.

Consider the function.

$$f(t) = \begin{cases} -10, & 0 < t < 1 \\ 0, & 1 < t < 2 \\ 10, & 2 < t < 3 \end{cases}$$

Step 2 of 2

Calculate the root mean square value of the function $f(t)$.

$$f_{rms} = \sqrt{\frac{1}{T} \int_0^T \{f(t)\}^2 dt}$$

Substitute 3 for T .

$$\begin{aligned} f_{rms} &= \sqrt{\frac{1}{3} \int_0^3 \{f(t)\}^2 dt} \\ &= \sqrt{\frac{1}{3} \left\{ \int_0^1 (-10)^2 dt + \int_1^2 (0)^2 dt + \int_2^3 (10)^2 dt \right\}} \\ &= \sqrt{\frac{1}{3} \left\{ \int_0^1 100 dt + 0 + \int_2^3 100 dt \right\}} \\ &= \sqrt{\frac{100}{3} \left\{ \int_0^1 dt + \int_2^3 dt \right\}} \\ f_{rms} &= \sqrt{\frac{100}{3} \{t|_0^1 + t|_2^3\}} \\ &= \sqrt{\frac{100}{3} \{(1-0) + (3-2)\}} \\ &= \sqrt{\frac{100}{3} (1+1)} \\ &= \sqrt{\frac{100}{3} (2)} \\ f_{rms} &= \sqrt{\frac{200}{3}} \\ &= \sqrt{66.667} \\ &= 8.165 \end{aligned}$$

Therefore, the root mean square value of the function $f(t)$ is, 8.165.